

Advanced Data Science Project Proposal

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Date: April 14, 2026

Project Overview:

For the Advanced Data Science Project, I will be working **solo**. I plan to explore delay propagations in airline networks. Commercial airlines run on a very tight schedule trying their best to promise consumers on time departures and arrivals. With crews, aircraft rotations, and weather there are many factors that can affect a single flight that as a result cascade downstream and can affect travel for the rest of the day or in some cases the rest of the week all around the US. I plan to model domestic airlines in the United States as a directed graph and apply various methods from CSCI 4022 to identify what airports and routes are the most influential to delays, how delays propagate, and what combinations of conditions are reliable to predict large scale cascades.

Data Source:

After some exploration I was able to find this [data](#) source which is observational from the Bureau of Transportation Statistics. This allows you to select specific statistics based on airline carriers on time performance. These features range from the airline, airport, city, destination, delay time, cancellation indicator, and so many more variables that are extremely valuable when attempting to create a model that can represent how delays move through the national airline system. I also am planning to join this data with hourly weather data from NOAA at major hub airports so I can factor in meteorological conditions. This is because the weather is one of the largest drivers of large-scale disruptions. It is also worth mentioning that the BTS data is collected through federal reporting requirements, rather than being scraped or experimental. To keep the scope of this project manageable, I plan to use data from the past 12-14 months. This should allow me to capture seasonal patterns.

The features that will be most impactful for this project will be scheduled and actual departure and arrival times. The origin and destinations will also be key factors. The carrier and delay causes may also help me with this model; time and day of the week may also be very impactful. I expect strong relationships between these variables. For example, late aircraft delays at a hub should correlate with downstream delays later in the day. I also expect to find carrier specific patterns because certain airlines concentrate their resources on specific hubs. I also expect peak hours and days of the week to expand everything.

Methods:

PageRank will be used to build a directed graph where the nodes are the airports, and the edges are flights weighted by how often delay at airport A is followed by a delay at airport B later that day. I will probably map this via flight numbers or tail numbers. PageRank scores will be able to tell me what airports are the influencers to large scale delays, essentially telling us what airport delays ripple outward.

Clustering (K-means) will be used to find structure in airport behavior. I will represent each airport as a feature vector. Then from there I plan to cluster the airports to find natural groupings. This will allow me to answer different questions like do clusters correspond to carrier hubs?

Motivation:

This summer I will be working for an airline; this makes this whole project directly relevant to the field I will be working in. An airline's operational reliability is what makes or breaks them in the eyes of a consumer. This will give me valuable insights going into my summer internship, being able to understand which airports and conditions most reliably trigger large cascades and maybe be able to leverage this information to suggest optimized routes in certain areas if we expect other airlines to face major delays. Prior work I have done in data mining has prepared me for the graph and clustering methods I will be using in this project.

Limitations:

One thing I have already noticed is that delay cause codes are self-reported by each airline and may be inconsistent as a result. Another challenge will be merging weather data with transportation data. But other than that, I have been provided with more than enough tools to create an informative model. The fact that I am using observational data limits my model to association and not causation. Even with these limitations I am proved with enough data and methods to build an informative model.

Links:

https://www.transtats.bts.gov/DL_SelectFields.aspx?QO_fu146_anzr=b0-gvzr&gnoyr_VQ=FGJ

https://transtats.bts.gov/DatabaselInfo.asp?QO_VQ=EFD